



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO VARUN FUEL AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

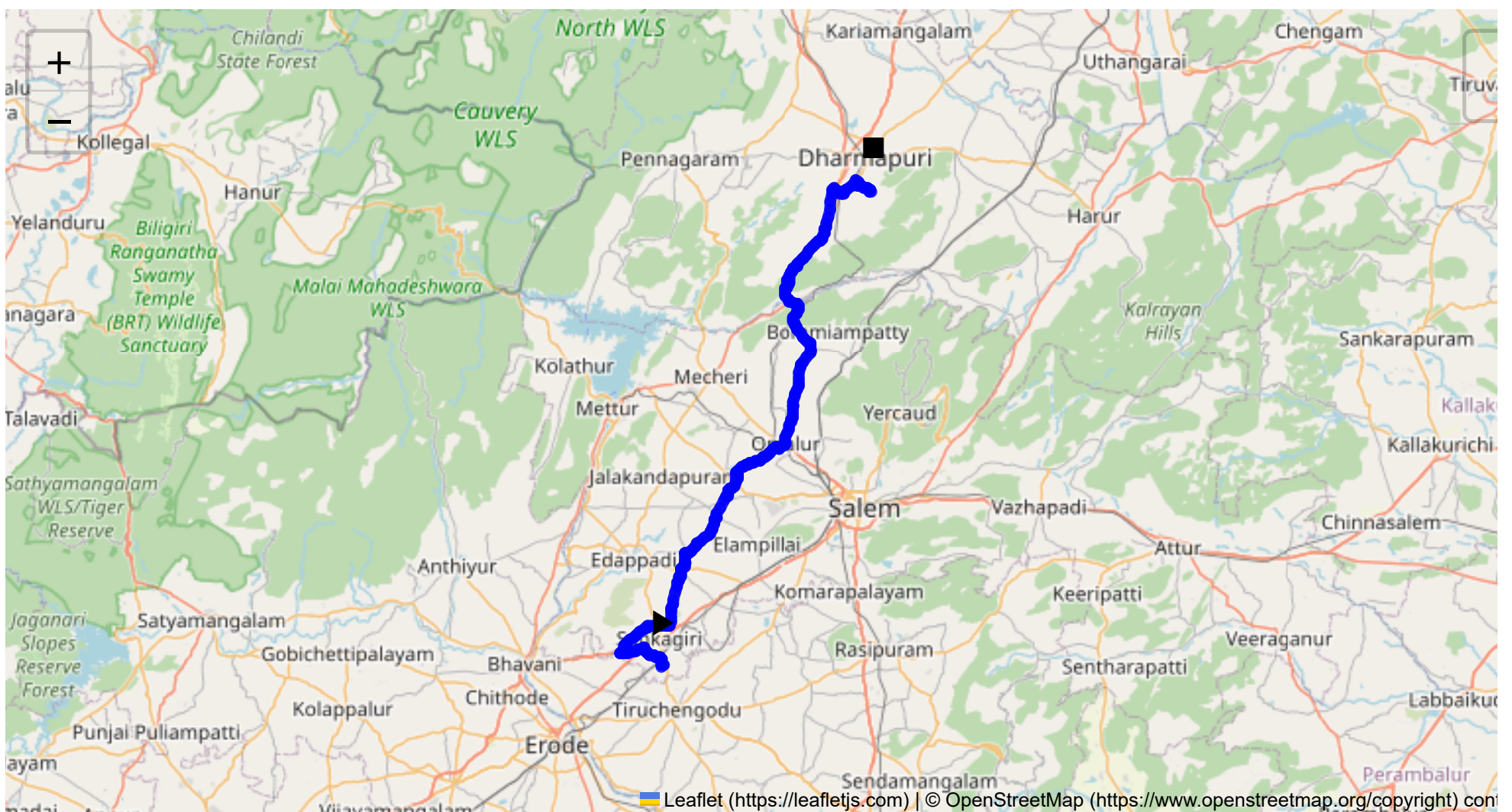
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 106.69 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.0 hours
Start: (11.4381, 77.8734)
End: (12.086688, 78.166412)



Welcome to the Journey Risk Management Study

Route Overview

The main route traverses multiple state highways and regional roads in Tamil Nadu, spanning various terrains and urban-rural landscapes. The total stretch is approximately 106.69 kilometers, often navigated by heavy vehicles, taking around 2.37 hours.

Weather Conditions

The region experiences a tropical climate, with high temperatures and humidity, particularly from March to June. Monsoon season from October to December can lead to heavy rainfall, potentially causing road flooding and landslides. Fog may occur during late nights and early mornings from December to February, affecting visibility.

Traffic Patterns

1. **Peak Hours:** Congestion is noticeable during early mornings (7-9 AM) and late afternoons (5-7 PM).
2. **Congested Areas:** Notably, Omalur and Salem city outskirts witness frequent traffic jams due to commercial activities and marketplaces.

Road Quality and Infrastructure

- Generally, the highways such as Salem-Kochi Hwy and NH 86 are in good condition, but connecting roads might be narrower and lesser maintained, especially in rural areas.
- Drive cautiously on narrower segments and unmarked curves.

Alternative Routes

In case of emergencies or blockades, rerouting via Salem-Attur or Salem-Tiruchengode highways, although longer, offer better infrastructure and safety.

Local Regulations on Hazardous Materials

Transport of hazardous materials needs to adhere to local and state transport laws, requiring appropriate permits and clearly marked vehicles. Nighttime transport may be restricted in certain municipalities.

Historical Incidents

While there are no recent major incidents reported in the public domain, minor accidents due to road conditions and weather are noted, mainly involving improper cargo management rather than the route itself.

Environmental Considerations

The route passes near agricultural zones and small water bodies; spillage or leaks can impact these sensitive areas. Certain zones may restrict access for heavy vehicles during specific harvest periods.

Communication Coverage

- **Dead Zones:** Remote stretches, especially between hills, might face connectivity issues. It is advisable to carry network-enhancing devices.

Emergency Response Times

- Response varies greatly across the route; urban areas like Salem offer rapid assistance (<15 minutes), whereas, in rural areas, it might extend up to 45 minutes.

Overall Summary

- **Risk Assessment:** The route presents moderate risk levels, manageable with proper precautions.
 - **Weather impact:** High in monsoon.
 - **Traffic congestion:** Manageable but requires planning around peak hours.
- Enhanced safety measures include planning for weather changes, avoiding peak times, ensuring up-to-date communication devices, and familiarizing with alternative routes and emergency contacts.

Drivers should remain vigilant and regularly monitor weather and traffic updates for a safer transit experience.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45543, 77.81740	30 KM/Hr
8	Turn	High	11.45631, 77.81668	15 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	Medium	11.56083, 77.89921	30 KM/Hr
11	Turn	Medium	11.56180, 77.90019	30 KM/Hr
12	Turn	Medium	11.58254, 77.90656	30 KM/Hr
13	Turn	Medium	11.64217, 77.95200	30 KM/Hr
14	Turn	Medium	11.64712, 77.95118	30 KM/Hr
15	Turn	High	11.67858, 77.96834	15 KM/Hr
16	Turn	High	11.70233, 77.97869	15 KM/Hr
17	Turn	Medium	11.71394, 77.99572	30 KM/Hr
18	Turn	Medium	11.71916, 78.01117	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
19	Turn	Medium	11.72559, 78.01758	30 KM/Hr
20	Turn	Medium	11.73787, 78.03528	30 KM/Hr
21	Turn	High	11.73651, 78.03868	15 KM/Hr
22	Turn	Medium	11.74120, 78.04588	30 KM/Hr
23	Turn	Medium	11.74257, 78.04708	30 KM/Hr
24	Turn	Medium	11.77998, 78.05786	30 KM/Hr
25	Turn	Medium	11.86915, 78.08382	30 KM/Hr
26	Turn	Medium	11.87998, 78.08170	30 KM/Hr
27	Turn	Medium	11.91266, 78.05852	30 KM/Hr
28	Turn	Medium	11.93746, 78.05331	30 KM/Hr
29	Turn	Medium	11.95531, 78.04698	30 KM/Hr
30	Turn	Medium	11.95803, 78.04964	30 KM/Hr
31	Turn	Medium	11.95854, 78.04990	30 KM/Hr
32	Turn	High	11.96015, 78.04945	15 KM/Hr
33	Turn	Medium	11.96370, 78.05142	30 KM/Hr
34	Turn	Medium	11.96433, 78.05113	30 KM/Hr
35	Turn	High	11.96542, 78.04938	15 KM/Hr
36	Turn	High	11.96650, 78.04927	15 KM/Hr
37	Turn	Medium	11.96804, 78.05080	30 KM/Hr
38	Blind Spot	Blind Spot	12.09201, 78.11462	10 KM/Hr
39	Turn	High	12.08762, 78.12236	15 KM/Hr
40	Turn	High	12.08688, 78.12234	15 KM/Hr
41	Turn	High	12.08464, 78.12765	15 KM/Hr
42	Turn	High	12.10313, 78.14568	15 KM/Hr
43	Turn	Medium	12.10181, 78.14798	30 KM/Hr
44	Turn	High	12.10183, 78.14823	15 KM/Hr
45	Turn	High	12.09968, 78.14988	15 KM/Hr
46	Turn	High	12.09652, 78.14894	15 KM/Hr
47	Blind Spot	Blind Spot	12.09165, 78.16520	10 KM/Hr
48	Turn	Medium	12.09093, 78.16513	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
8	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
9	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	clinic	S.K. Clinic	12.0969793, 78.1400279	30 km/h	Medium
31	hospital	Sri Sai Sakthi Siddha Hospital	12.0965297, 78.1396218	30 km/h	Medium
32	clinic	Sri Maruthi Clinic and Scans	12.095525, 78.1386931	30 km/h	Medium
33	clinic	Apollo Homeo Clinic	12.104278, 78.147159	30 km/h	Medium
34	clinic	Homoeopathy & Herbal Clinic	12.1040673, 78.1471714	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56083, 77.89921



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56180, 77.90019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58254, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70233, 77.97869



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71394, 77.99572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71916, 78.01117



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72559, 78.01758



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73787, 78.03528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74120, 78.04588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74257, 78.04708



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77998, 78.05786



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86915, 78.08382



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87998, 78.08170



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91266, 78.05852



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95803, 78.04964



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96015, 78.04945

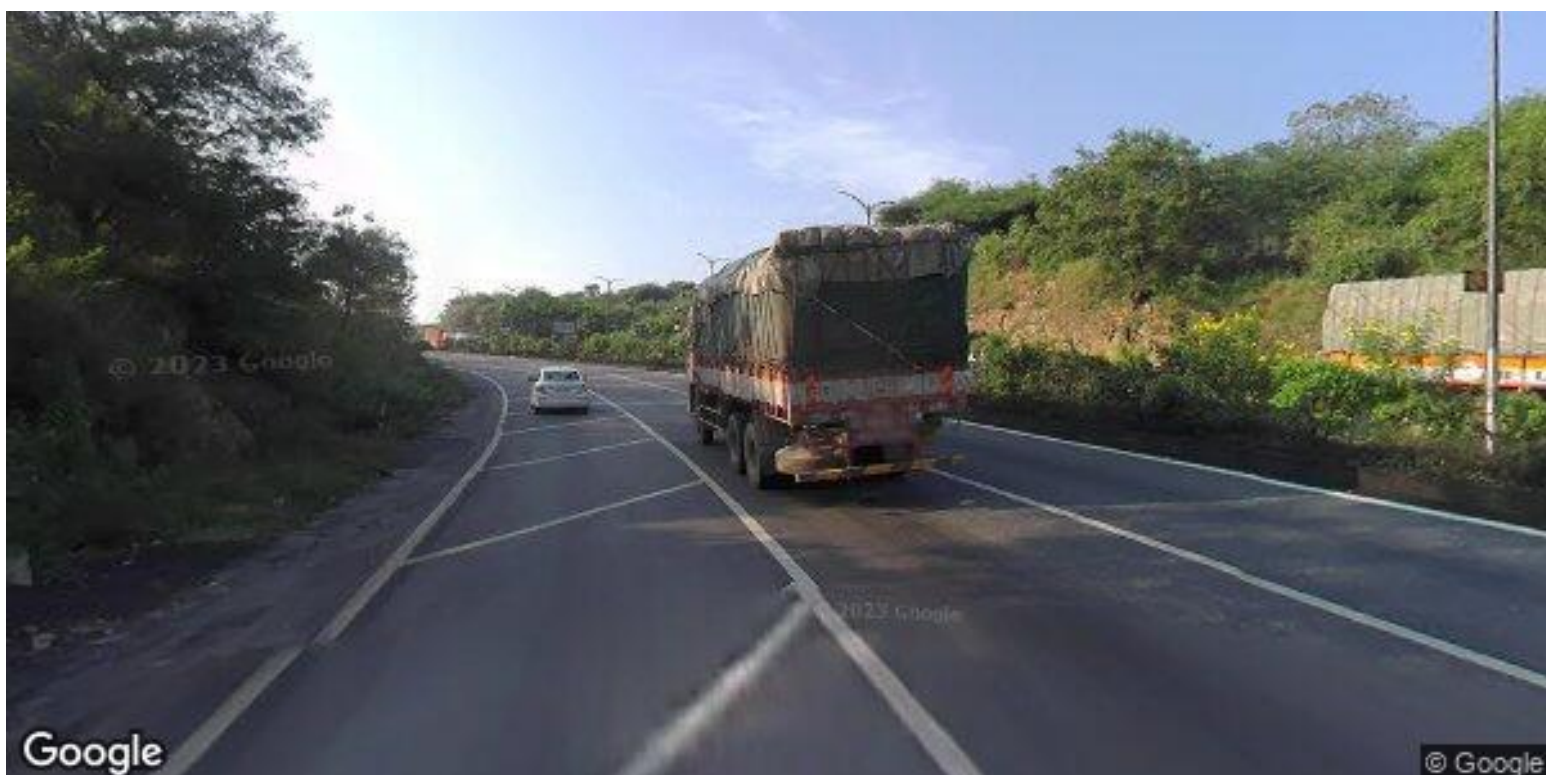


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96804, 78.05080



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.09201, 78.11462



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.08762, 78.12236



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.08688, 78.12234



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.08464, 78.12765



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.10313, 78.14568

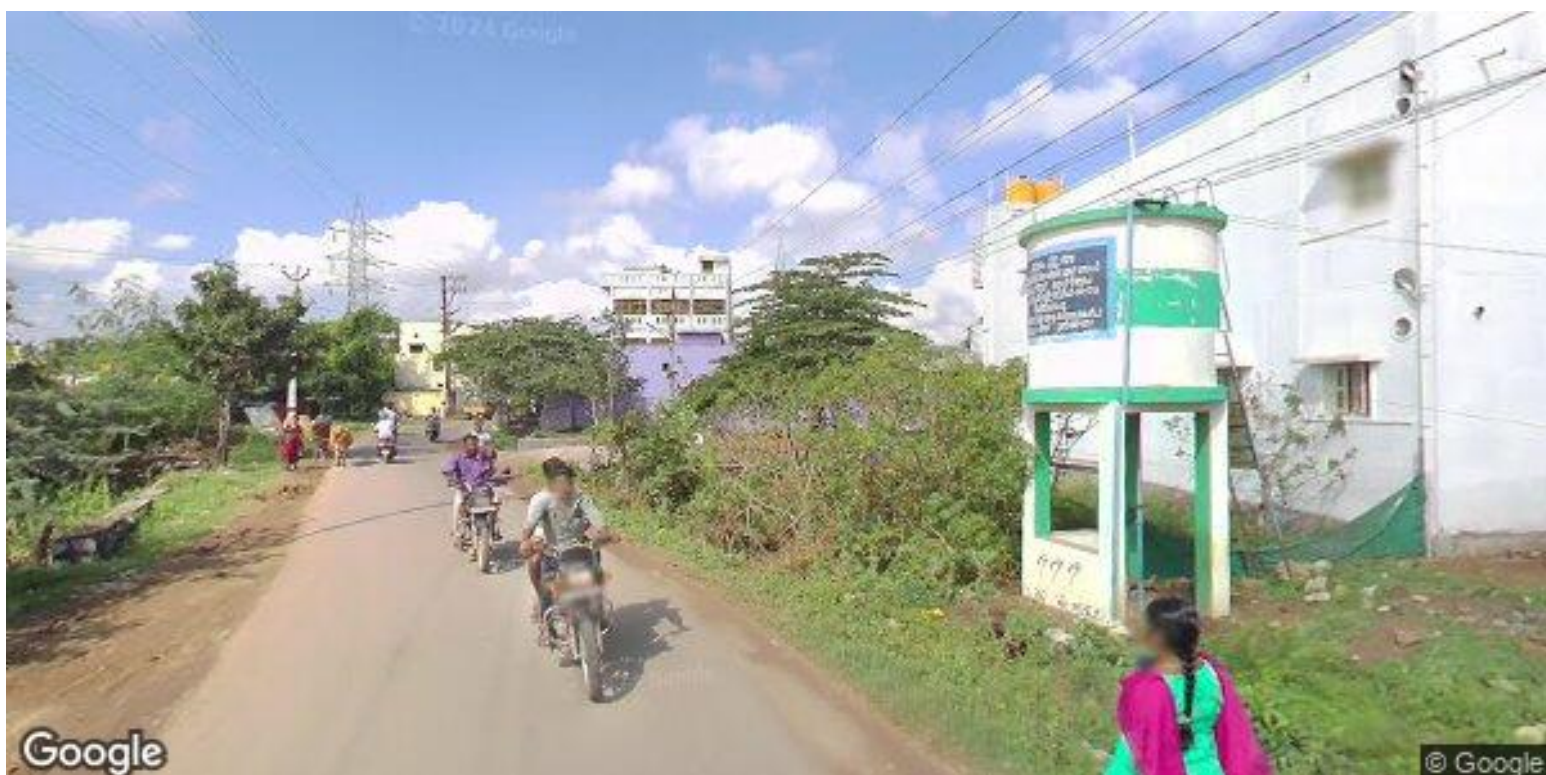


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.10181, 78.14798



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.10183, 78.14823



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.09968, 78.14988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.09652, 78.14894



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.09165, 78.16520



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.09093, 78.16513
